

PROBLEMS

- 5-1. In the network of the figure, the switch K is closed at $t = 0$ with the capacitor uncharged. Find values for i , di/dt and d^2i/dt^2 at $t = 0+$, for element values as follows: $V = 100$ V, $R = 1000\ \Omega$, and $C = 1\ \mu\text{F}$.

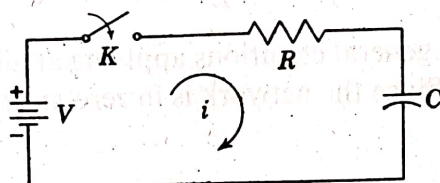


Fig. P5-1.

- 5-2. In the given network, K is closed at $t = 0$ with zero current in the inductor. Find the values of i , di/dt , and d^2i/dt^2 at $t = 0+$ if $R = 10\ \Omega$, $L = 1$ H, and $V = 100$ V.

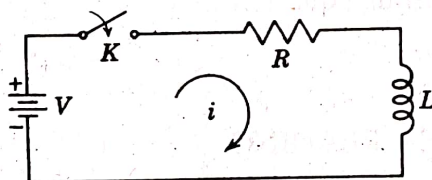


Fig. P5-2.

- 5-3. In the network of the figure, K is changed from position a to b at $t = 0$. Solve for i , di/dt , and d^2i/dt^2 at $t = 0+$ if $R = 1000\ \Omega$, $L = 1$ H, $C = 0.1\ \mu\text{F}$, and $V = 100$ V.

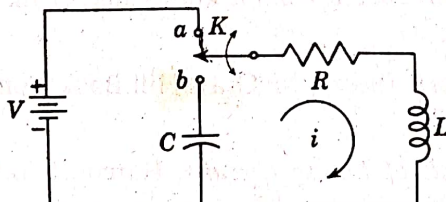


Fig. P5-3.

- 5-4. For the network and the conditions stated in Prob. 4-3, determine the values of dv_1/dt and dv_2/dt at $t = 0+$.
- 5-5. For the network described in Prob. 4-7, determine values of d^2v_2/dt^2 and d^3v_2/dt^3 at $t = 0+$.
- 5-6. The network shown in the accompanying figure is in the steady state with the switch K closed. At $t = 0$, the switch is opened. Determine the voltage across the switch, v_K , and dv_K/dt at $t = 0+$.

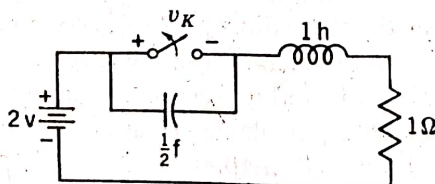


Fig. P5-6.

- 5-7. In the given network, the switch K is opened at $t = 0$. At $t = 0+$, solve for the values of v , dv/dt , and d^2v/dt^2 if $I = 10$ amp, $R = 1000\ \Omega$, and $C = 1\ \mu\text{F}$.

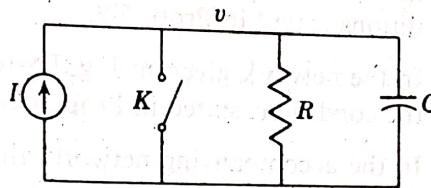


Fig. P5-7.

- 5-8. The network shown in the figure has the switch K opened at $t = 0$. Solve for v , dv/dt , and d^2v/dt^2 at $t = 0+$ if $I = 1$ amp, $R = 100\ \Omega$, and $L = 1$ H.

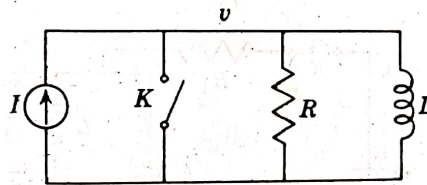


Fig. P5-8.

- 5-9. In the network shown in the figure, a steady state is reached with the switch K open. At $t = 0$, the switch is closed. For the element values given, determine the value of $v_a(0-)$ and $v_a(0+)$.

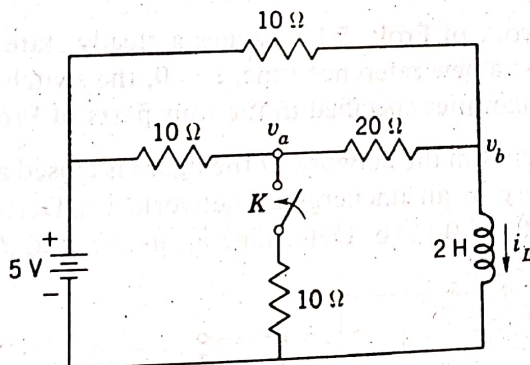


Fig. P5-9.

- 5-10. In the accompanying figure is shown a network in which a steady state is reached with switch K open. At $t = 0$, the switch is closed.

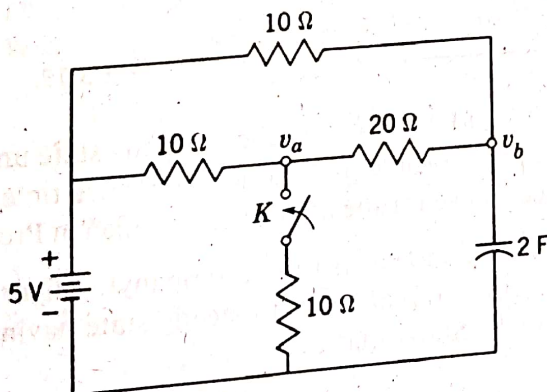


Fig. P5-10.

For the element values given, determine the values of $v_a(0-)$ and $v_a(0+)$.

5-11. In the network of Fig. P5-9, determine $i_L(0+)$ and $i_L(\infty)$ for the conditions stated in Prob. 5-9.

5-12. In the network given in Fig. P5-10, determine $v_b(0+)$ and $v_b(\infty)$ for the conditions stated in Prob. 5-10.

5-13. In the accompanying network, the switch K is closed at $t = 0$ with zero capacitor voltage and zero inductor current. Solve for (a) v_1 and v_2 at $t = 0+$, (b) v_1 and v_2 at $t = \infty$, (c) dv_1/dt and dv_2/dt at $t = 0+$, (d) d^2v_2/dt^2 at $t = 0+$.

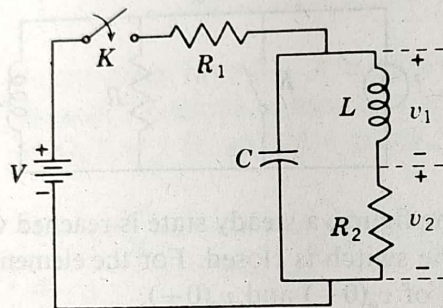


Fig. P5-13.

5-14. The network of Prob. 5-13 reaches a steady state with the switch K closed. At a new reference time, $t = 0$, the switch K is opened. Solve for the quantities specified in the four parts of Prob. 5-13.

5-15. The switch K in the network of the figure is closed at $t = 0$ connecting the battery to an unenergized network. (a) Determine i , di/dt , and d^2i/dt^2 at $t = 0+$. (b) Determine v_1 , dv_1/dt , and d^2v_1/dt^2 at $t = 0+$.

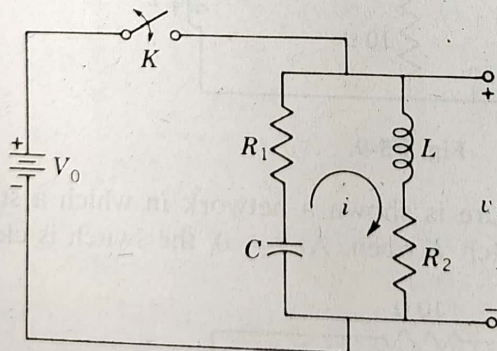


Fig. P5-15.

5-16. The network of Prob. 5-15 reaches a steady state under the conditions specified in that problem. At a new reference time, $t = 0$, the switch K is opened. Solve for the quantities specified in Prob. 5-15 at $t = 0+$.

5-17. In the network shown in the accompanying figure, the switch K is changed from a to b at $t = 0$ (a steady state having been established at position a). Show that at $t = 0+$,

$$i_1 = i_2 = -\frac{V}{R_1 + R_2 + R_3}, \quad i_3 = 0$$

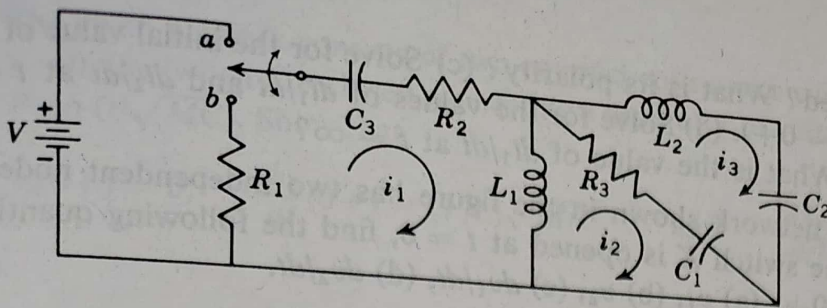


Fig. P5-17.

- 5-18. In the given network, the capacitor C_1 is charged to voltage V_0 and the switch K is closed at $t = 0$. When $R_1 = 2 \text{ M}\Omega$, $V_0 = 1000 \text{ V}$, $R_2 = 1 \text{ M}\Omega$, $C_1 = 10 \mu\text{F}$, and $C_2 = 20 \mu\text{F}$, solve for d^2i_2/dt^2 at $t = 0+$.

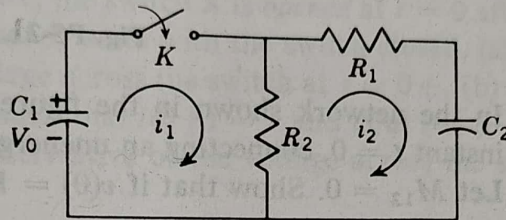


Fig. P5-18.

- 5-19. In the circuit shown in the figure, the switch K is closed at $t = 0$ connecting a voltage, $V_0 \sin \omega t$, to the parallel RL - RC circuit. Find (a) di_1/dt and (b) di_2/dt at $t = 0+$.

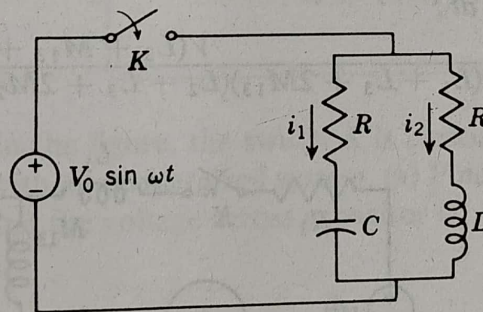


Fig. P5-19.

- 5-20. In the network shown, a steady state is reached with the switch K open with $V = 100 \text{ V}$, $R_1 = 10 \Omega$, $R_2 = 20 \Omega$, $R_3 = 20 \Omega$, $L = 1 \text{ H}$, and $C = 1 \mu\text{F}$. At time $t = 0$, the switch is closed. (a) Write the integrodifferential equations for the network after the switch is closed. (b) What is the voltage V_0 across C before the switch is closed.

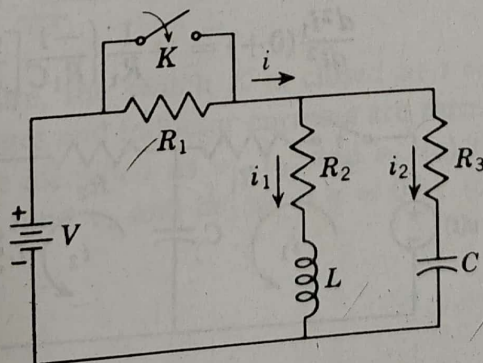


Fig. P5-20.

closed? What is its polarity? (c) Solve for the initial value of i_1 and $i_2(t = 0+)$. (d) Solve for the values of di_1/dt and di_2/dt at $t = 0+$. (e) What is the value of di_1/dt at $t = \infty$?

- 5-21. The network shown in the figure has two independent node pairs. If the switch K is opened at $t = 0$, find the following quantities at $t = 0+$: (a) v_1 , (b) v_2 , (c) dv_1/dt , (d) dv_2/dt .

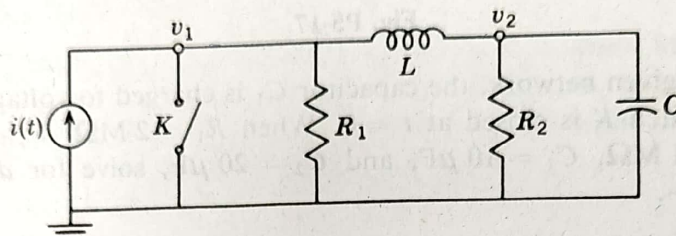


Fig. P5-21.

- 5-22. In the network shown in the figure, the switch K is closed at the instant $t = 0$, connecting an unenergized system to a voltage source. Let $M_{12} = 0$. Show that if $v(0) = V$, then:

$$\begin{aligned} \frac{di_1}{dt}(0+) &= \frac{V(L_2 + L_3 + 2M_{23})}{(L_1 + L_3 + 2M_{13})(L_2 + L_3 + 2M_{23}) - (L_3 + M_{13} + M_{23})^2} \\ \frac{di_2}{dt}(0+) &= \frac{V(L_3 + M_{13} + M_{23})}{(L_1 + L_3 + 2M_{13})(L_2 + L_3 + 2M_{23}) - (L_3 + M_{13} + M_{23})^2} \end{aligned}$$

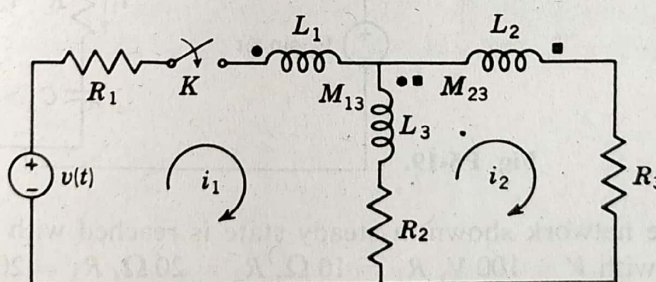


Fig. P5-22.

- 5-23. For the network of the figure, show that if K is closed at $t = 0$,

$$\frac{d^2 i_1}{dt^2}(0+) = -\frac{1}{R_1} \left\{ \frac{-1}{R_1 C} \left[\frac{v(0)}{R_1 C} - \frac{dv}{dt}(0) \right] - \frac{d^2 v}{dt^2}(0) \right\}$$

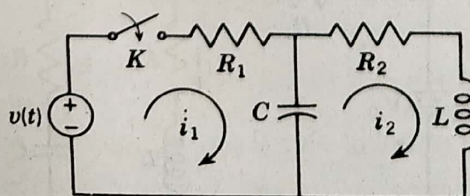


Fig. P5-23.

- 5-24. The given network consists of two coupled coils and a capacitor. At $t = 0$, the switch K is closed connecting a generator of voltage, $v(t) = V \sin(t/\sqrt{MC})$. Show that

$$v_a(0+) = 0, \quad \frac{dv_a}{dt}(0+) = (V/L)\sqrt{M/C}, \quad \text{and} \quad \frac{d^2v_a}{dt^2}(0+) = 0$$

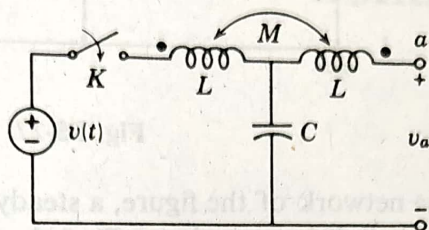


Fig. P5-24.

- 5-25. In the network of the figure, the switch K is opened at $t = 0$ after the network has attained a steady state with the switch closed. (a) Find an expression for the voltage across the switch at $t = 0+$. (b) If the parameters are adjusted such that $i(0+) = 1$ and $di/dt(0+) = -1$, what is the value of the derivative of the voltage across the switch, $dv_K/dt(0+)$?

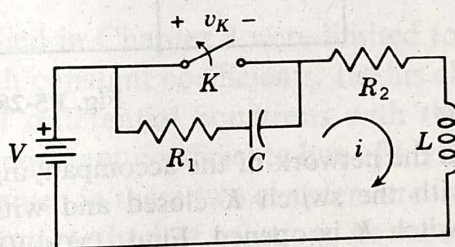


Fig. P5-25.

- 5-26. In the network shown in the figure, the switch K is closed at $t = 0$ connecting the battery with an unenergized system. (a) Find the voltage v_a at $t = 0+$. (b) Find the voltage across capacitor C_1 at $t = \infty$.

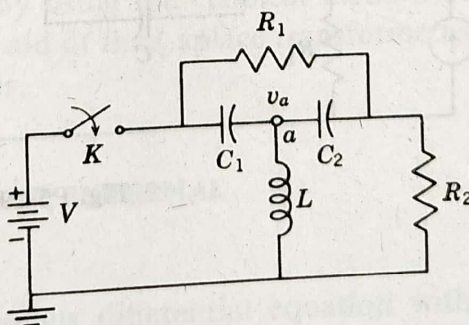


Fig. P5-26.

- 5-27. In the network of the figure, the switch K is closed at $t = 0$. At $t = 0-$, all capacitor voltages and inductor currents are zero. Three node-to-datum voltages are identified as v_1 , v_2 , and v_3 . (a) Find v_1 and dv_1/dt at $t = 0+$. (b) Find v_2 and dv_2/dt at $t = 0+$. (c) Find v_3 and dv_3/dt at $t = 0+$.

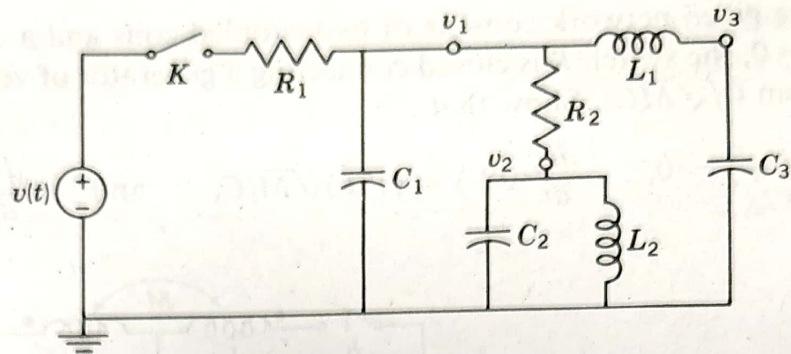


Fig. P5-27.

- 5-28. In the network of the figure, a steady state is reached, and at $t = 0$, the switch K is opened. (a) Find the voltage across the switch, v_K at $t = 0+$. (b) Find dv_K/dt at $t = 0+$.

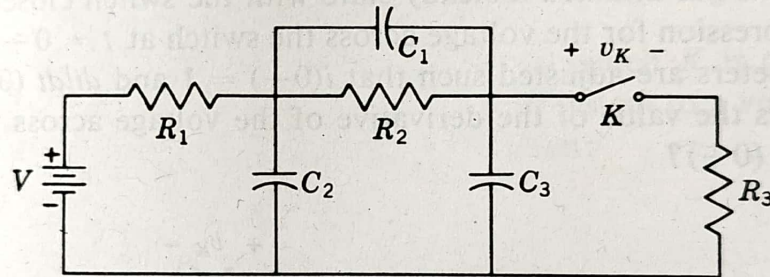


Fig. P5-28.

- 5-29. In the network of the accompanying figure, a steady state is reached with the switch K closed and with $i = I_0$, a constant. At $t = 0$, switch K is opened. Find: (a) $v_2(0-)$, (b) $v_2(0+)$, and (c) $(dv_2/dt)(0+)$.

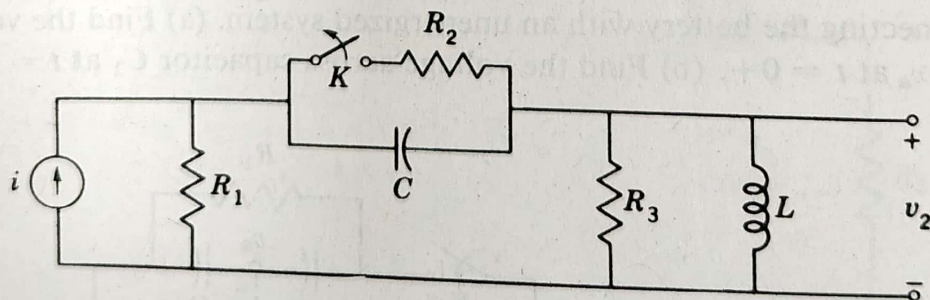


Fig. P5-29.

$R_2 - R_2 e^{-R_1 t/L}$. 4-13. (b) $i = (Vt/L)e^{-R_1 t/L}$. 4-16. (h)

$$i = \frac{[(a-1) \sin 2t - 2 \cos t]e^{-t} + 2e^{-at}}{(a-1)^2 + 4}$$

4-17. $i_L = 0.6 + 0.067e^{-3.57t}$. 4-20. (a) $R = 10^4 \Omega$, (b) $C = 2.5 \mu\text{F}$.

4-21. (b) $i = 0.01(1 - e^{-16t})$ amp. 4-22. $w_C/w_T = (\frac{1}{2})(1 - e^{-t/RC})$.

Chapter 5

5-1. 0.1, -100, 100,000. 5-3. $d^2i/dt^2(0+) = -9 \times 10^5$ amp/sec.² 5-5. -8 V/sec², 26 V/sec³. 5-6. 0 V/sec². 5-8. 100, -10⁴, 10⁶. 5-13. (b) 0, $R_2 V/(R_1 + R_2)$, (c) V/CR_1 , 0, (d) $R_2 V/R_1 LC$. 5-15. $d^2v_1/dt^2(0+) = V_0/R$. 5-18. 1.406×10^{-5} amp/sec.² 5-20. (b) 66.7 V, (c) 3.33, 1.67 amps, (d) 33.3, -83,300. 5-21. $R_1 i(0+)$, 0, $R_1[(di/dt)(0+) - Ri(0+)/L]$, 0. 5-25. (a) VR_1/R_2 , (b) $(1/C) - R_1$. 5-27. (a) 0, $v(0+)/R_1 C_1$, (b) 0, 0, (c) 0, 0. 5-28. (a) $v_K(0+) = VR_3/(R_1 + R_2 + R_3)$.

Chapter 6

6-3. (f) $i = (A \cos \sqrt{1.75}t + B \sin \sqrt{1.75}t)e^{-t/2}$. (h) $i = Ae^{-2t} + Bte^{-2t}$. 6-5. (a) $i = 2e^{-t} - e^{-2t}$. (b) $i = 3e^{-2t} - 2e^{-3t}$. (f) $i = e^{-0.5t}(\cos 1.32t + 0.379 \sin 1.32t)$. (h) $i = e^{-2t} + te^{-2t}$. 6-9. $i = Ae^{-4.18t} + e^{-1.12t}(B \cos 1.06t + C \sin 1.06t)$. 6-11. $t_1 = 1/\alpha$. 6-14. $i = e^{-140t} \sin 1570t$. 6-15. $i_L = 10 \cos 316t$. 6-18. (b) $i = K_1 e^{-t} + K_2 e^{-2t} - \frac{15}{4} + (5t/2)$, (e) $v = (K_1 + t)e^{-2t} + (K_2 - 5t)e^{-3t}$. 6-23. $i = 5t^2 e^{-t}$. 6-24. $i = 10^{-4} e^{-t}(\cos t - \sin t) - 10^{-4} e^{-10t}$. 6-28. $i = 10^{-3}(20.3 \cos 377t + 14.8 \sin 1414t - 0.8 \cos 1414t)$ amp. 6-29. $s(500) = -250 \pm j968$, $s(1000) = -500 \pm j866$, $s(3000) = -2,620, -380$, $s(5000) = -4,790, -210$. 6-31. Characteristic equation: $(5 - K_1)s^2 + (6 - 2K_1)s + (2 - K_1) = 0$; the system is stable for $K_1 < 2$. 6-33. (d) $w_S = w_R + w_L = (V^2/R)[t + (L/R)(e^{-Rt/L} - 1)]$.

Chapter 7

7-9. $F_1(s) = 2/[s(s^2 + 4)]$. 7-16. $F_8(s) = s^2/[(s^2 + a^2)^2]$. 7-22. $(V/R_1)e^{-(R_1+R_2)t/L}$. 7-24. $(V/R) \cos(t/\sqrt{LC})$. 7-30. (a) $K_0 = \frac{3}{8}$, $K_2 = \frac{1}{4}$, $K_4 = \frac{3}{8}$. 7-35. $i = K_1 e^t + K_2 e^{-t} - 25 + (\frac{1}{3})e^{2t}$. 7-37. $i = (\frac{1}{3})t^3 + 2e^{-t} + 2$. 7-42. $v_a = 13(e^{-0.0215t} - e^{-0.1285t})$. 7-43. 16.7 sec. 7-44. 14 V in 29.2 sec. 7-45. (a) $3.33 + 1.21e^{-6.35t} - 4.5e^{-23.6t}$. 7-46. $1020(e^{-0.155t} - e^{-0.645t})$. 7-49. $v_a = 4 + e^{-0.75t}(-1.5 \cos 0.25t + 0.5 \sin 0.25t)$. 7-50. $i_2 = -5 - 1.34e^{-0.707t} + 16.3e^{0.707t}$.

Chapter 8

8-1. $u(t-2) - u(t-3) + u(t-7) - u(t-8)$. 8-9. $v = K_0[t^2 u(t) - 4(t-1)u(t-1) - (t-2)^2 u(t-2)]$. 8-10. Partial answer: $v = be^{(\ln(b/a)t/c} \cdot u(t) - \dots$. 8-14. (c) $i = [5 \times 10^5 t - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t) - [5 \times 10^5(t-10^{-6}) - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t-10^{-6})$. 8-17. $i(t) = [1 - e^{-(t-1)}]u(t-1) + [1 - e^{-(t-2)}]u(t-2) + [1 - e^{-(t-3)}]u(t-3) + [1 - e^{-(t-4)}]u(t-4) - 4[1 - e^{-(t-5)}]u(t-5)$. 8-19. $F_1(s) = K[(1/s^2) + (t_0/s)]$. 8-26. $V(s) = (1/s)[e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} - 4e^{-5s}]$. 8-31. $F(s) = 1/[s(1 - e^{-s})]$. 8-33. $F(s) = \{1/[(s + \frac{1}{2})(1 + e^{-(0.5+s)})]\}$. 8-34. (a) $f(t) = te^{at}$.